

ROSHAN KUMAR SINGH

Full Stack Developer (MERN) · AI-Integrated Applications · Open to Internship & Entry-Level Roles
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PROFILE SUMMARY

Third-year B.Tech CSE student with hands-on experience building and deploying full-stack web applications using the MERN stack and FastAPI, with applied AI/ML integration (LLMs, voice AI, reinforcement learning). Advanced to the Grand Finale of Scaler's AI Hackathon (top 100 of 30,000+ teams) and the finals of Meta's PyTorch OpenEnv Hackathon (top 800 of 31,000+ teams). Focused on building accessible, civic-impact technology. Seeking internship or entry-level full-stack/frontend roles.

EDUCATION

B.Tech, Computer Science & Engineering — AKS University, Satna, MP

Sep 2023 – Oct 2027

CGPA: 7.44 (current)

EXPERIENCE

Freelance MERN Stack Developer — Self-Employed

Sep 2025 – Present

- Designed and shipped multiple full-stack web applications end-to-end (wireframe through deployment) using React, Node.js/Express, and MongoDB/PostgreSQL.
- Built and deployed an AI-powered civic platform integrating LLM APIs; the project advanced to the Grand Finale of Scaler's AI Hackathon (top 100 of 30,000+ teams).
- Managed cloud deployments on AWS, Vercel, and Render, including environment configuration, API security (JWT), and database hosting.

PROJECTS

Vibely — Real-Time Social Platform [Live](#) | [GitHub](#)

- Built a production-style social platform with real-time messaging and typing indicators (Socket.io), media uploads, and dual authentication (Google OAuth + JWT).
- Built UI with Next.js, TypeScript, shadcn/ui, and Framer Motion; deployed frontend on Vercel and backend on Render with MongoDB.

FleetWatch — Multi-Agent Fraud Detection (RL) · Team Project [Live](#) | [GitHub](#)

- Collaborated on a reinforcement-learning environment for multi-agent fleet fraud detection with 5 progressive curriculum tasks (GPS tampering to 3-agent collusion), built for Meta's PyTorch OpenEnv Hackathon.
- Contributed to a 7-signal anti-gaming reward system and LoRA fine-tuning of Llama-3-8B (~30 min on a T4 GPU), achieving an 11x improvement in mean reward over baseline. Reached finals (top 800 of 31,000+ teams), Bangalore.

AKS Beauty Hut — E-Commerce Platform (Freelance Client) [Live](#) | [GitHub](#)

- Delivered a responsive MERN e-commerce site for a local retail business, including product catalog, cart/checkout, user authentication, and an admin inventory panel.
- Deployed via Vercel with a mobile-first design approach to support the client's customer base.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python

Frontend: React.js, Next.js, Tailwind CSS, shadcn/ui, Framer Motion, HTML5, CSS3

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, Socket.io

Databases: MongoDB, PostgreSQL, MySQL, Redis, DynamoDB, Pinecone

AI/ML: TensorFlow, PyTorch, OpenAI API, Hugging Face, LangChain, LoRA Fine-Tuning, Reinforcement Learning

Cloud & Tools: AWS (Bedrock, Amplify), Docker, Kubernetes, Firebase, Vercel, Render, Git/GitHub, GitHub Actions

HACKATHONS & ACHIEVEMENTS

- Scaler AI Hackathon — Grand Finalist (top 100 of 30,000+ teams) for an AI-powered civic platform with LLM integration.
- Meta PyTorch OpenEnv Hackathon — Finalist (top 800 of 31,000+ teams); presented project in Bangalore, Apr 2026.
- Smart India Hackathon (SIH) 2025 — Team project addressing a civic-issue-reporting use case.

CERTIFICATIONS

- Software Developer (IT-ITES Sector) — Skill India Mission (PMKVY), NCVET-certified, 420-hour program (2025).
- Web Development (HTML5, CSS3, JS ES6, React.js, MongoDB) — AKS University (2024).
- Web Development Completion Certificate — IBM SkillBuild; Full MERN Stack Training (Sigma Batch 8.0) — Apna College.